

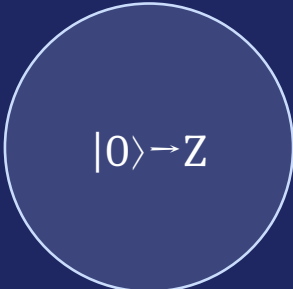
LLM-Guided VQC Design: From Pilot to Rigorous Benchmark

Amplitude encoding, quantum readout, and budget-matched evaluation of quantum architecture search (QAS)

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ML4SCI / Google Summer of Code 2026 — weekly update

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$|0\rangle \rightarrow Z$

Does LLM guidance actually help quantum architecture search?

“Does LLM guidance improve VQC architecture search over random, evolutionary, greedy, and reference ansätze under the same candidate-evaluation budget?”

1 Prior agentic-VQC work motivates the direction

Knipfer et al. 2026 and Sakka et al. 2026 show LLM agents can design VQCs — but with no budget-matched non-LLM baselines and no seed-replicated statistics.

2 Anytime improvement alone is uninterpretable

Random search also improves over iterations; only equal-budget comparison isolates the value of the proposal strategy.

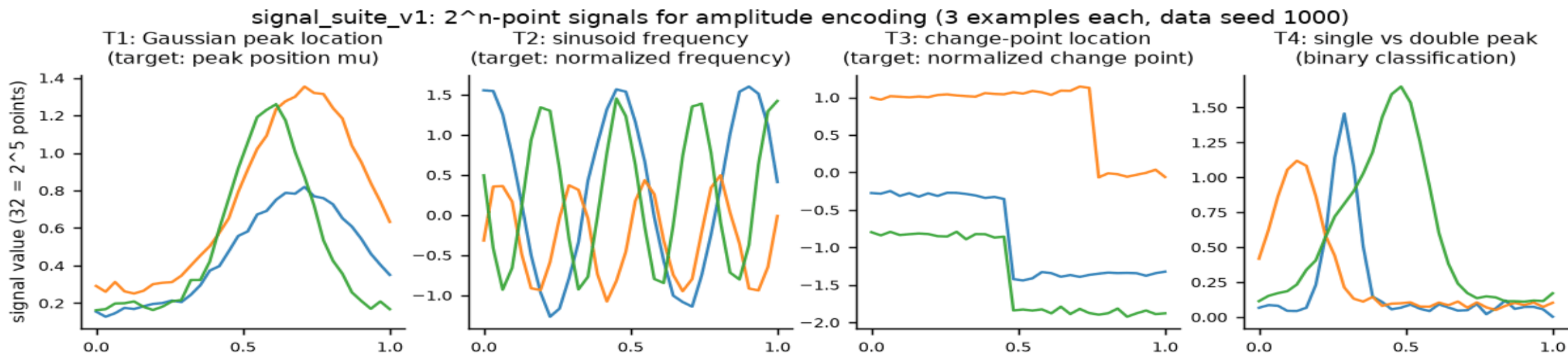
3 Our contribution: the controlled evaluation layer

Same typed circuit grammar, same evaluator, same unique-candidate budget, paired seeds, protected test — for every arm, LLM or classical.

From a 2-seed pilot to a preregistered benchmark

	July pilot (now: E0 provenance only)	bench_v2 (this benchmark)
Replication	2 seeds	10 paired replicates (5 for scaling), paired seeds per arm
Budget	B = 4 proposals	B = 16–24 unique candidate evaluations, exact ledgers
Tasks	Gaussian peak only (n=3)	signal_suite T1–T4; scaling $n \in \{3,4,6,8\}$
Method scope	joint structure+ θ only	Track A (structure-only, shared trainer) + Track B (joint, verbatim θ)
Baselines	random only	random, evolutionary, greedy, 4 fixed reference ansätze, joint-evolutionary
Claims	descriptive (integration demo)	preregistered RQ1–RQ6, Holm-corrected paired tests, effect sizes
Integrity	protected test, 2 committed seeds	+ resource metrics, machine-checked 17-criterion goal contract, break-tests

Four controlled signal tasks, sized for amplitude encoding



2^n features

Every task generates exactly 2^n -point signals — the amplitude vector of n qubits. No padding, no truncation.

Nuisance randomization

Amplitude, phase, baseline, widths, noise randomized so no fixed convention leaks the label.

Scaling profiles

T1/T2 frozen for $n \in \{3, 4, 6, 8\}$ (E3, running); T3/T4 at $n=5$. Frequency cap $N/4$ keeps T2 below Nyquist.

Splits

256 train / 256 val / 2048 protected test per replicate; deterministic hashes; disjointness enforced.

Two tracks separate architecture quality from angle luck

Track A — architecture search (primary)

- Arms propose structure only, in one shared layered grammar
- One shared AdamW inner loop trains θ (40 epochs, frozen for every arm)
- θ -seed = $f(\text{task, data seed, search seed, structural hash})$ — same circuit $\Rightarrow$ same result in every arm (global cache)
- Feedback to searchers: validation metric only

Track B — direct joint proposal (ablation)

- Candidates carry gates, wires, AND numeric angles
- θ evaluated verbatim — structurally NO optimizer on this path (AST-enforced)
- Candidate identity includes θ : same structure + new angles = new candidate
- Mandatory classical rival: mixed discrete/continuous evolutionary search

Why it matters: a joint proposer can look good by guessing lucky angles rather than good architectures. Only the split design attributes credit correctly — E5 additionally re-optimizes θ on frozen LLM architectures at exactly matched objective-call budgets.

Evaluation hygiene is enforced by construction, then attacked



Validation-only feedback

SearchFeedback has no field that could hold a test value; arms never see test data.



Paired isolation

data seed 1000+r, search seed 2000+r shared across arms; θ -seeds derived, never global.



Machine-checked completion

17-criterion goal contract; checker currently fails honestly (matrix incomplete) and rejects removed or interrupted cells.



Protected test, once

“Protected-test RMSE” = test metric of the validation-selected candidate, evaluated once after selection freeze — never feedback.



Crash-resumable stores

Per-cell SQLite + budget ledger; interrupt-and-resume proven equal to an uninterrupted run.



Break-tests

Deliberately broke 3 safeguards: 2 exposed real blind spots (now closed with stronger tests). No secrets or raw transcripts in artifacts.

Completed: E0 provenance replay and E1 classical joint search

E0 — pilot replay

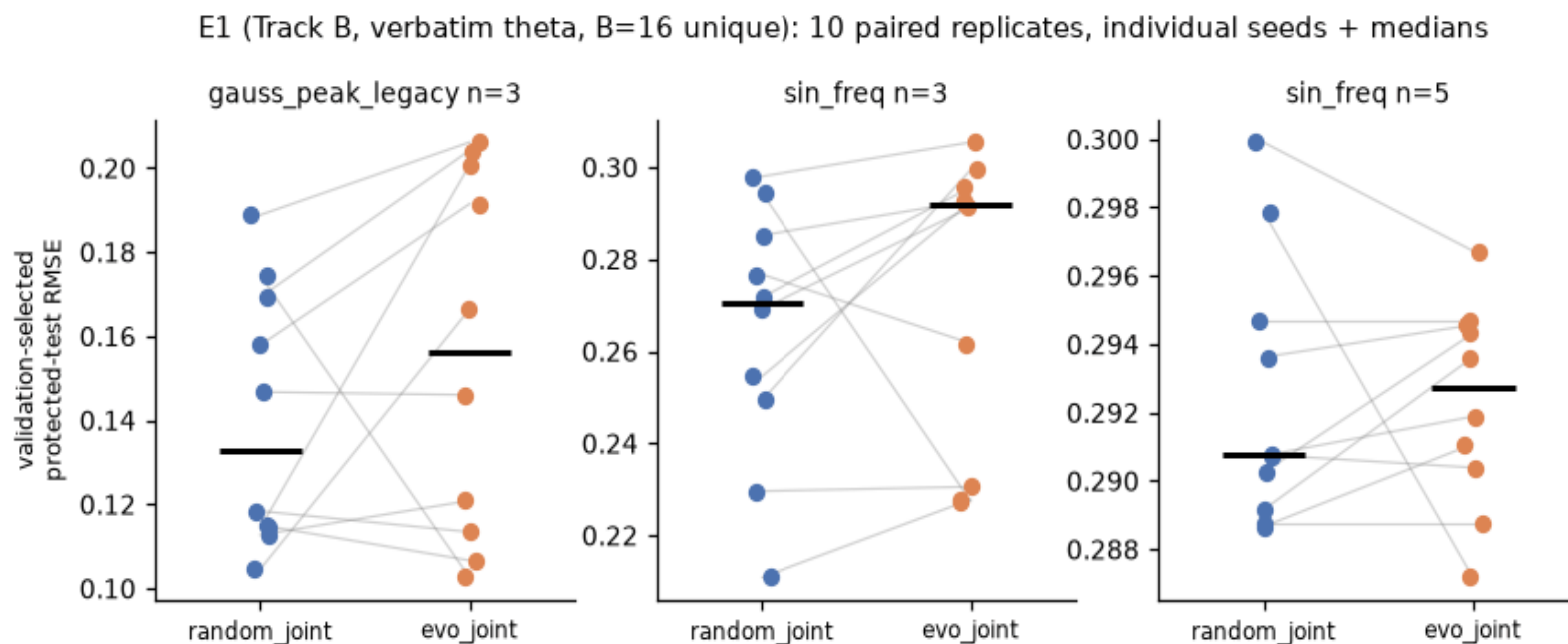
All 21 committed real-LLM pilot candidates re-evaluated offline from stored operations + θ .

Exact match

validation AND test values, all 3 arms, both seeds

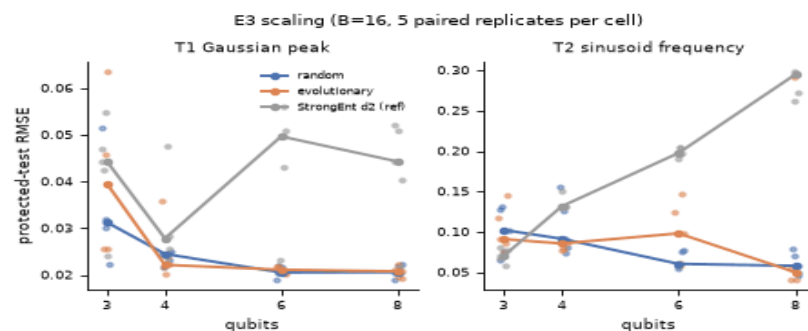
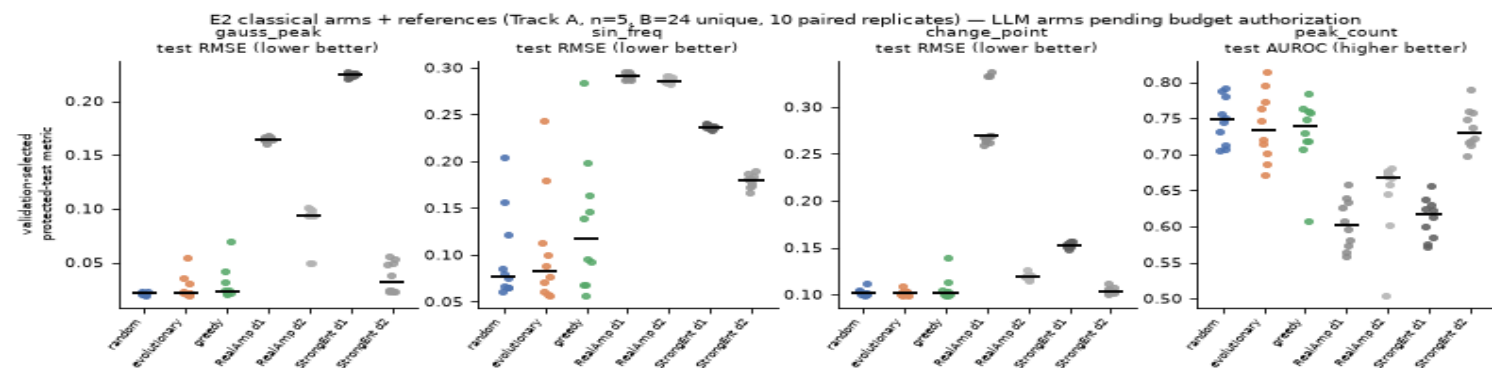
0 new API calls

The July numbers are provenance-verified — and remain a 2-seed integration demo, not evidence.



E1 (B=16, verbatim θ , 10 paired replicates): `random_joint` and `evolutionary_joint` are statistically close — no Holm-corrected difference (all $p \geq 0.28$). T2 at $n=5$ appears difficult for both: test RMSE ≈ 0.29 for every replicate, near the target-scale spread — verbatim- θ proposals rarely beat it at this budget. LLM joint arms: pending budget authorization.

Classical matrix complete: E2 at n=5, E3 scaling to 8 qubits

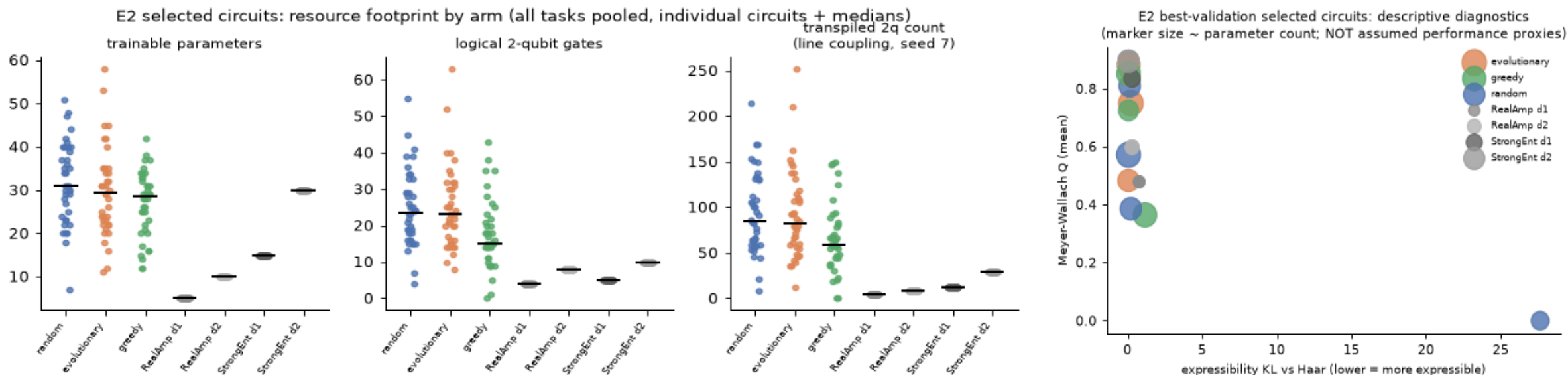


Findings

- n=5, 10 paired reps: searched circuits beat the shallow reference on regression tasks (T1 random vs StrongEnt-d2, $p_{\text{Holm}} = 0.032$).
- random / evolutionary / greedy stay statistically indistinguishable from each other on every task.
- Scaling (descriptive): on T2 the reference degrades with n (median RMSE $0.070 \rightarrow 0.295$ from n=3 to n=8) while searched arms improve ($0.102 \rightarrow 0.050$). All 5/5 paired replicates favour search at n=8, but with 5 reps the smallest attainable paired p is 0.0625, so no Holm-corrected significance is claimed.
- **This is the bar any LLM arm must beat.**

460 / 460 classical cells complete, 0 failures (E1 60 · E2 280 · E3 120) | LLM 0 / 220 blocked | E4 / E5 pending

Selected circuits: resources measured, diagnostics descriptive



Accuracy is bought with hardware cost: searched circuits use ~3–10× the transpiled 2-qubit gates of the fixed references (line coupling, fixed seed) — the Pareto view is part of the final analysis. **Diagnostics characterize circuits; they are not assumed to predict task performance** (Sim et al. framing; RQ5 tests the association empirically, with circuit-size controls). Expressibility KL + Meyer–Wallach Q computed for the 28 best-validation selected circuits. Shot-noise and noisy-simulator robustness (E4): pending — runs after the full matrix.

Next steps

- Classical matrix is done: 460 / 460 cells, 0 failures, n=8 actually simulated
- Run the real-LLM matrix (220 cells, both tracks) once budget is authorized
- Complete dependent analyses: E5 θ -isolation, E4 shot/noise robustness
- Regenerate final paper-style figures + statistical report from stores (goal checker must exit 0)
- Next mentor update: full budget-matched LLM-vs-classical comparison

Decision needed — API budget authorization

LLM_API_BUDGET_USD is not set, so all real-LLM cells are blocked by policy: an API key alone is not spending authorization, and a Claude Max subscription does not authorize experiment API spending.

Suggested cap 120 (conservative \$0.05/call reservation, hard 40-call/cell cap); expected actual cost $\approx$ \$5–20 on a mini-class model. One-line resume command in docs/research/BLOCKED.md.